

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, X: $\tilde{A}_2$ COMPLEXES, HIGHER SIMPLICIAL LAPLACIANS, AND THE FATE OF THE TRACE OBSTRUCTION IN RANK TWO

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ABSTRACT. The series has so far lived on quotients of the Bruhat–Tits tree of PGL_2 . This paper moves one rank up, to finite $\tilde{A}_2$ complexes — quotients of two-dimensional buildings of order q , locally the building of PGL_3 over a local field with residue field $\mathbb{F}_q$ — and asks which of the rank-one theorems survive. Three things are established. First, the Hodge Laplacians $\Delta_0, \Delta_1, \Delta_2$ of the typed chain complex give higher critical groups $\mathrm{Jac}_i(Y) = \mathrm{Tor\,coker}\,\Delta_i$, computed exactly on thin tori and on five thick complexes of order 2 with 21, 24, 42, 84 and 168 vertices, built here from a triangle presentation over $\mathrm{PG}(2, 2)$ compatible with a Singer cycle and verified to have Heawood-graph links; the Hodge vertex Laplacian is no longer a Hecke operator, and the arithmetic bridge of [I, Thm. 3.5] is carried instead by the *Hecke–Laplacian* $\Delta^H = (q^3 - 1)I + A_1 - qA_2$, the value at $u = 1$ of the cubic pencil $P(u) = \det(I - uA_1 + qu^2A_2 - q^3u^3I)$, whose critical group has order $\frac{1}{n} \prod_j |P_j(1)|$, a product of inverse local GL_3 L -values. Second, the companion $C^{(3)}$ of the cubic pencil is unimodularly equivalent to $\Delta^H \oplus I \oplus I$ exactly as in [II], but the trace obstruction of [II] does *not* transfer: the $\mathbb{Z}/3$ type grading forces $\mathrm{Tr}(C^{(3)})^k = 0$ for $3 \nmid k$, and a block computation gives $\mathrm{Tr}(C^{(3)})^3 = n(q - 1)^2(q + 1) \geq 0$. There is no cubic trace obstruction; on the five thick complexes, all of which turn out to be Ramanujan, every net trace of the reduced (type-preserving) pencil is non-negative and its Perron condition holds, so by the theorem of Kim–Ormes–Roush the nonzero spectrum of $C^{(3)}$ is that of a non-negative integer matrix, irreducible of period three — the rank-two pencil is spectrally realizable, the opposite of the rank-one situation, and the realization problem for the PGL_3 Hecke critical group is open and unobstructed. Third, the shadow construction of [VI] extends to arbitrary integer matrices with positive diagonal by a signed-shadow lemma in which positive off-diagonal entries are produced by one-directional auxiliary vertices with double loops; it yields unital Kirchberg algebras $\mathcal{O}(Y, \Delta_1)$ and $\mathcal{O}(Y, \Delta^H)$ with $K_0 \cong \mathbb{Z}^{b_1} \oplus \mathrm{Jac}_1(Y)$ and $K_0 \cong \mathbb{Z} \oplus \mathrm{Jac}^H(Y)$ respectively, functorially in the complex. What the specification predicted and what is proved differ in two places, and the paper says where.

STATUS OF THE RESULTS

Every numbered statement is proved. The inputs are finite linear algebra, the classical Cuntz–Krieger facts quoted in [VI, VIII], the Cartwright–Mantero–Steger–Zappa theorem that a triangle presentation gives a group acting simply transitively on the vertices of an $\tilde{A}_2$ building [1], and, for one corollary that nothing else depends on, the Kim–Ormes–Roush theorem on the nonzero spectra of primitive integer matrices [4]. The Kang–Li zeta function of PGL_3 complexes [3] is context for the cubic pencil, not an input. Every finite identity was machine-verified exactly (Table 1); the five thick complexes are constructed by the script, and their local structure (Heawood links, regularity, flag condition) is checked rather than assumed — with the result that the flag condition holds only

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Paper X of the series; Papers I–IX are cited as [I]–[IX]. The verification script `pgl3.building.py` in the ancillary bundle constructs every complex used and reproduces every computational claim; its output is archived as `pgl3.building.output.txt`.

for the largest of them, a distinction the theorems respect. Section 7 records what is open and the corrections to the specification.

1. $\tilde{A}_2$ COMPLEXES OF ORDER q

Definition 1.1 ($\tilde{A}_2$ complex). An $\tilde{A}_2$ complex of order $q \geq 1$ is a finite connected pure two-dimensional simplicial complex Y with vertex set V ($n := |V|$), edge set E and triangle set F , together with a type function $\tau: V \rightarrow \mathbb{Z}/3$, such that

- (i) every triangle has one vertex of each type;
- (ii) every vertex has exactly $q^2 + q + 1$ neighbours of type $\tau(v) + 1$ and $q^2 + q + 1$ of type $\tau(v) + 2$;
- (iii) every edge lies in exactly $q + 1$ triangles;
- (iv) (flag condition) every closed walk $x \rightarrow y \rightarrow z \rightarrow x$ with $\tau(y) = \tau(x) + 1$, $\tau(z) = \tau(x) + 2$ is the boundary of a triangle.

Every edge is oriented from type t to type $t + 1$ and every triangle (a, b, c) is listed with types $(t, t + 1, t + 2)$, so that $\partial_2(a, b, c) = (a, b) + (b, c) + (c, a)$ and $\partial_1(a, b) = b - a$ are canonical; $A_1 \in M_n(\{0, 1\})$ is the adjacency $x \rightarrow y$ for edges, $A_2 := A_1^t$, and we assume $A_1 A_2 = A_2 A_1$ (true for building quotients, where A_1, A_2 generate the commutative spherical Hecke algebra; verified on every complex used).

Quotients of a thick $\tilde{A}_2$ building of order $q \geq 2$ by a type-rotating, freely acting group with injectivity radius ≥ 2 are $\tilde{A}_2$ complexes in this sense, with vertex links the incidence graph of $\text{PG}(2, q)$; the flag condition holds because buildings are flag complexes. Thin complexes ($q = 1$) are triangulated flat tori (Construction 1.3).

Lemma 1.2 (Counts). $|E| = n(q^2 + q + 1)$, $|F| = \frac{1}{3}n(q^2 + q + 1)(q + 1)$, $\text{Tr } A_1^3 = 3|F|$, $\text{Tr}(A_1 A_2) = |E|$, and $\chi(Y) = n(1 + \frac{1}{3}(q^2 + q + 1)(q - 2))$. For $q = 2$: $|E| = |F| = 7n$ and $\chi(Y) = n$.

Proof. (ii) gives $|E|$; each vertex lies in $(q^2 + q + 1)(q + 1)$ triangles by (ii), (iii), each counted three times; $\text{Tr } A_1^3$ counts closed type-increasing 3-walks, three per triangle by (iv); $\text{Tr}(A_1 A_1^t) = \sum_{x, y} A_1(x, y)^2 = |E|$. $\square$

Construction 1.3 (Thin tori). On $\mathbb{Z}^2$ let $d_1 = (1, 0)$, $d_2 = (0, 1)$, $d_3 = (-1, -1)$, edges $v \rightarrow v + d_i$, type $\tau(a, b) = a + b \bmod 3$, triangles $\{v, v + d_i, v + d_i + d_j\}$, $i \neq j$. This is the $\tilde{A}_2$ Coxeter complex (one apartment); its quotient by $m\mathbb{Z}^2$, $3 \mid m$, $m \geq 3$, is an $\tilde{A}_2$ complex of order 1 with m^2 vertices, $b_1 = 2$, $b_2 = 1$, hexagonal links.

Construction 1.4 (Thick complexes of order 2). Label the points of $\text{PG}(2, 2)$ by $\mathbb{Z}/7$ with lines $\{i, i + 1, i + 3\}$ and let $\lambda(x) = \{x + 1, x + 2, x + 4\}$, the Singer cycle. A triangle presentation compatible with λ [1] is a set T of 21 triples with (A) $(x, y, z) \in T \Rightarrow (y, z, x) \in T$, (B) $(x, y, \cdot) \in T$ iff $y \in \lambda(x)$, (C) the third entry is determined by the first two. Exhaustive search over all bijections without absolute points finds, for λ the Singer cycle, the presentation with cyclic orbits

$$T/\mathbb{Z}_3 = \{(0, 1, 3), (0, 2, 6), (0, 4, 5), (1, 2, 4), (1, 5, 6), (2, 3, 5), (3, 4, 6)\}$$

(axioms re-verified by the script). By [1] the group $\Gamma = \langle a_0, \dots, a_6 \mid a_x a_y a_z = 1, (x, y, z) \in T \rangle$ acts simply transitively and type-rotatingly on the vertices of an $\tilde{A}_2$ building of order 2; its type-preserving subgroup Γ^0 has index 3, and $X_3 := \Gamma^0 \backslash B$ is the Δ -complex with three vertices, 21 edges and 21 triangles. We compute $H_1(X_3; \mathbb{Z}) \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/14$ and build the abelian voltage covers of X_3 with deck group A for the quotients $A = \mathbb{Z}/7$, $(\mathbb{Z}/2)^3$, $\mathbb{Z}/14$ and $H_1(X_3)$ itself, exactly as the derived graphs of [II, §5] in one dimension higher: vertices (i, g) , edges lifted by voltages in A , triangles lifting because their boundaries are null-homologous. The results $Y_{21}, Y_{24}, Y_{42}, Y_{84}, Y_{168}$ are simplicial, every vertex link is the Heawood graph (bipartite, 3-regular, 14 vertices, girth 6), and conditions (i)–(iii) hold, with $|E| = |F| = 7n$ (Table 1, line (P)). The flag condition (iv) holds for

Y_{168} and fails for the four smaller covers: there the deck group is too small, some product $a_x a_y a_{z'}$ with $(x, y, z') \notin T$ has trivial voltage, and $\text{Tr } A_1^3$ exceeds $3|F|$ by $28n, 21n, 12n, 4n$ respectively. So (iv) is not implied by the link condition, and the five complexes separate the two hypotheses cleanly: Y_{168} is an $\tilde{A}_2$ complex in the full sense of Definition 1.1, the others satisfy (i)–(iii) only. The count $168 = |\text{PGL}_3(\mathbb{F}_2)|$ for the maximal abelian cover suggests that Y_{168} is a congruence quotient; we neither need nor claim this. Whether the building is that of $\text{PGL}_3(\mathbb{F}_2((t)))$, of $\text{PGL}_3(\mathbb{Q}_2)$, or exotic is not decided here: every statement below depends only on the local structure of Definition 1.1, and the statements that use (iv) are marked.

2. HODGE LAPLACIANS AND HIGHER CRITICAL GROUPS

Definition 2.1. $\Delta_0 := \partial_1 \partial_1^t$ on C_0 , $\Delta_1 := \partial_1^t \partial_1 + \partial_2 \partial_2^t$ on C_1 , $\Delta_2 := \partial_2^t \partial_2$ on C_2 , and $\text{Jac}_i(Y) := \text{Tor coker}_{\mathbb{Z}} \Delta_i$, the i -th critical group. Following Duval–Klivans–Martin [2] we also record $K(Y) := \ker \partial_1 / \text{im}(\partial_2 \partial_2^t)$.

Proposition 2.2 (Structure). $\partial_1 \partial_2 = 0$; $\text{coker } \Delta_i \cong \mathbb{Z}^{b_i} \oplus \text{Jac}_i(Y)$ with b_i the Betti numbers of Y ; $b_0 = 1$; $\Delta_0 = 2(q^2 + q + 1)I - (A_1 + A_2)$ is the Laplacian of the 1-skeleton, so $\text{Jac}_0(Y)$ is its sandpile group with $|\text{Jac}_0(Y)|$ the number of spanning trees of the 1-skeleton; Δ_1 has diagonal entries $q + 3$, entry 0 at pairs of edges spanning a triangle, and entries ± 1 at pairs sharing a vertex and no triangle.

Proof. $\partial_1 \partial_2(a, b, c) = (b - a) + (c - b) + (a - c) = 0$. Each Δ_i is symmetric with $\ker \Delta_i = \ker \partial_i \cap \ker \partial_{i+1}^t$ of rank b_i (Hodge), so its cokernel has free rank b_i . The diagonal of Δ_1 is $2 + (q + 1)$ by (iii); for edges $e = (a, b)$, $f = (b, c)$ in a triangle the down-term $\langle \partial e, \partial f \rangle = -1$ cancels the up-term $+1$, and without a triangle only the down-term survives. $\square$

3. THE HECKE–LAPLACIAN AND THE GL_3 L -VALUES

Definition 3.1 (Cubic pencil and Hecke–Laplacian). $P(u) := I - uA_1 + qu^2A_2 - q^3u^3I \in M_n(\mathbb{Z}[u])$ and

$$\Delta^H := -P(1) = (q^3 - 1)I + A_1 - qA_2, \quad \text{Jac}^H(Y) := \text{Tor coker}_{\mathbb{Z}} \Delta^H.$$

For a joint eigenpair (a_j, b_j) of (A_1, A_2) write $P_j(u) = 1 - a_ju + qb_ju^2 - q^3u^3 = \prod_{i=1}^3(1 - \lambda_{j,i}u)$; the trivial pair $a = b = q^2 + q + 1$ gives $(1 - u)(1 - qu)(1 - q^2u)$.

The pencil is the vertex factor of the Kang–Li zeta function of a PGL_3 complex [3], and $P_j(u)^{-1}$ is the local L -factor of the unramified GL_3 Hecke eigenform with Satake parameters $\lambda_{j,i}/q$; we use only the polynomial.

Theorem 3.2 (Hecke critical group and L -values). (i) $\Delta^H \mathbf{1} = 0$ and $\mathbf{1}^t \Delta^H = 0$.

(ii) If $P_j(1) \neq 0$ for every nontrivial j (no nontrivial eigenform has a Satake parameter $\lambda_{j,i} = 1$; automatic when $|\lambda_{j,i}| = q$), then $\text{rk } \Delta^H = n - 1$, all n principal cofactors of Δ^H are equal, and

$$\text{coker}_{\mathbb{Z}} \Delta^H \cong \mathbb{Z} \oplus \text{Jac}^H(Y), \quad |\text{Jac}^H(Y)| = \frac{1}{n} \prod_{j \text{ nontrivial}} |P_j(1)|.$$

(iii) $\Delta^H \neq \Delta_0$ for $q \geq 1$, and Δ_0 is not a specialization of the pencil: no u gives $P(u) \in \mathbb{Q}^\times \Delta_0$. For PGL_2 the two operators coincide $((q+1)I - T_p = \Delta_0 = -P(1)$ with $P(u) = I - uT_p + qu^2$).

Proof. (i) Row sums: $(q^3 - 1) + (q^2 + q + 1) - q(q^2 + q + 1) = 0$; column sums likewise since $\mathbf{1}^t A_1 = \mathbf{1}^t A_2 = (q^2 + q + 1)\mathbf{1}^t$. (ii) The eigenvalues of Δ^H on the joint eigenspaces are $-P_j(1)$, zero exactly for the trivial pair under the hypothesis, so the kernel is $\mathbb{Z}\mathbf{1}$ on both sides; the equal-cofactor argument of [III, proof of Thm. 2.1, Step 2] applies verbatim to a matrix with two-sided kernel $\mathbf{1}$ and gives $\prod_{\mu \neq 0} \mu = n\kappa$ with κ the common cofactor $= |\text{Tor coker}|$. (iii) $P(u) \in \mathbb{Q}\Delta_0$ would need

$-u : qu^2 = -1 : -1$, i.e. $u = -1/q$, where $P(-1/q) = 2I + (A_1 + A_2)/q$ has the wrong constant term unless $2q = 2(q^2 + q + 1)$, impossible. $\square$

Theorem 3.3 (Companion reduction). *Let $C^{(3)} = \begin{pmatrix} A_1 & -qA_2 & q^3I \\ I & 0 & 0 \\ 0 & I & 0 \end{pmatrix}$, so that $\det(I - uC^{(3)}) = \det P(u)$. With $U = \begin{pmatrix} I & q^3I - qA_2 & q^3I \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix}$ and $V = \begin{pmatrix} I & 0 & 0 \\ I & I & 0 \\ I & I & I \end{pmatrix}$, both unimodular,*

$$U(I - C^{(3)})V = P(1) \oplus I_n \oplus I_n = (-\Delta^H) \oplus I_n \oplus I_n, \quad \text{hence} \quad \text{coker}_{\mathbb{Z}}(I - C^{(3)}) \cong \mathbb{Z} \oplus \text{Jac}^H(Y).$$

Proof. $I - C^{(3)} = \begin{pmatrix} I - A_1 & qA_2 & -q^3I \\ -I & I & 0 \\ 0 & -I & I \end{pmatrix}$; the first block row of $U(I - C^{(3)})$ is $(I - A_1, qA_2, -q^3I) + (q^3I - qA_2)(-I, I, 0) + q^3(0, -I, I) = (P(1), 0, 0)$, and the column operations V clear the subdiagonal $-I$'s. This is [II, §1.1] one degree higher. $\square$

4. THE FATE OF THE TRACE OBSTRUCTION

Theorem 4.1 (Grading). $\text{Tr}(A_1^\alpha A_2^\beta) = 0$ unless $\alpha \equiv \beta \pmod{3}$. Consequently $\text{Tr}(C^{(3)})^k = 0$ for every k with $3 \nmid k$, and the spectrum of $C^{(3)}$ is invariant under multiplication by cube roots of unity.

Proof. A closed walk contributing to the trace changes type by $\alpha + 2\beta \equiv \alpha - \beta$. $\text{Tr}(C^{(3)})^k = \sum_j p_k(a_j, b_j)$ with p_k the k -th power sum of the roots of $\lambda^3 - a\lambda^2 + qb\lambda - q^3$, a polynomial in monomials $a^\alpha (qb)^\beta (q^3)^\gamma$ with $\alpha + 2\beta + 3\gamma = k$, so $\alpha \equiv \beta$ forces $3 \mid k$. Equivalently $D_\omega = \text{diag}(\omega^{\tau(v)}) \oplus \dots$ conjugates $\omega C^{(3)}$ to $C^{(3)}$. $\square$

Theorem 4.2 (No cubic trace obstruction). *For every $\tilde{A}_2$ complex of order q ,*

$$\text{Tr } C^{(3)} = 0, \quad \text{Tr}(C^{(3)})^2 = 0, \quad \text{Tr}(C^{(3)})^3 = 3|F| - 3q|E| + 3nq^3 = n(q-1)^2(q+1) \geq 0,$$

with equality only in the thin case $q = 1$. Hence the mechanism of [II, Thms. 1.5–1.6] — a negative trace of a power of the companion, incompatible with any non-negative matrix of any size — has no analogue in rank two at orders ≤ 5 : the first net trace that is not forced to be non-negative by these identities is tr_6 .

Proof. $p_1 = a$, $p_2 = a^2 - 2qb$, $p_3 = a^3 - 3qab + 3q^3$ (Newton). $\text{Tr } A_1 = 0$ (no loops), $\text{Tr } A_1^2 = 0$ and $\text{Tr } A_2 = 0$ by Theorem 4.1, $\text{Tr } A_1^3 = 3|F|$ and $\text{Tr}(A_1 A_2) = |E|$ by Lemma 1.2; substitute the counts: $(q^2 + q + 1)(q + 1) - 3q(q^2 + q + 1) + 3q^3 = q^3 - q^2 - q + 1 = (q - 1)^2(q + 1)$. The net trace $\text{tr}_m = \sum_{d|m} \mu(m/d) \text{Tr}(C^{(3)})^d$ vanishes for $m \leq 5$, $m \neq 3$, and equals $\text{Tr}(C^{(3)})^3 \geq 0$ for $m = 3$. Without the flag condition the identity reads $\text{Tr}(C^{(3)})^3 = \text{Tr } A_1^3 - 3q|E| + 3nq^3$, still non-negative since $\text{Tr } A_1^3 \geq 3|F|$: on Y_{168} the value is $504 = n(q - 1)^2(q + 1)$ exactly; on Y_{21} it is $651 = 63 + 588$, the excess being the spurious closed 3-walks (Table 1). $\square$

Remark 4.3 (Why rank one is different). In [II] the companion of $1 - uT_p + qu^2$ has $\text{Tr } C^2 = \text{Tr } T_p^2 - 2qn = n(q + 1) - 2qn < 0$: the $\mathbb{Z}/2$ type structure of a graph allows a walk to return in two steps (edge reversal), so the symmetric adjacency contributes $\text{Tr } T_p^2 = 2|E| > 0$ but not enough to beat the $-2qn$ of the negative block. In rank two the type grading is $\mathbb{Z}/3$: a type-increasing walk cannot return before three steps, the negative block $-qA_2$ is invisible to $\text{Tr}(C^{(3)})^k$ for $k < 3$, and at $k = 3$ the chambers contribute $3|F| = n(q^2 + q + 1)(q + 1)$, which exceeds the $3q|E| - 3nq^3 = n(q^2 + q + 1) \cdot 3q - 3nq^3$ removed by the negative block. The obstruction of [II] is a rank-one accident of parity.

Definition 4.4 (Reduced pencil). By Theorem 4.1, $\det P(u) \in \mathbb{Z}[u^3]$; write $\det P(u) = Q(u^3)$ with $Q \in \mathbb{Z}[v]$ of degree n . Its roots are λ^3 , one for each ω -orbit $\{\lambda, \omega\lambda, \omega^2\lambda\}$ of eigenvalues of $C^{(3)}$: the trivial orbits give $1, q^3, q^6$, and the power sums of the multiset Λ_Q of roots are $p_d(\Lambda_Q) = \frac{1}{3} \text{Tr}(C^{(3)})^{3d}$. Λ_Q is the spectrum of the type-preserving (three-step) dynamics.

Theorem 4.5 (Spectral realizability of the thick examples). *For each of $Y_{21}, Y_{24}, Y_{42}, Y_{84}, Y_{168}$:*

- (i) $C^{(3)}$ has exactly nine eigenvalues of modulus $\geq q$: the orbits of $1, q, q^2$ under ω ; every other eigenvalue has modulus exactly q (numeric, to 10^{-6}): the complexes are Ramanujan in the sense of [3]. In particular $C^{(3)}$ itself never satisfies the Perron condition of a primitive matrix — $q^2, q^2\omega, q^2\omega^2$ are all dominant — which is forced by the grading and is the signature of an irreducible matrix of period 3.
- (ii) For Λ_Q : q^6 is simple and strictly dominant; the net traces

$$\mathrm{tr}_m(\Lambda_Q) = \sum_{d|m} \mu(m/d) \frac{1}{3} \mathrm{Tr}(C^{(3)})^{3d}$$

are non-negative for $m \leq 10$ (exact integers) and provably positive for $m \geq m_0 \in \{2, 3\}$ (Lemma 4.6); hence by [4] there is a primitive non-negative integer matrix B_0 with nonzero spectrum Λ_Q .

- (iii) Consequently the block-cyclic non-negative integer matrix $\begin{pmatrix} 0 & B_0 & 0 \\ 0 & 0 & I \\ I & 0 & 0 \end{pmatrix}$ has the nonzero spectrum of $C^{(3)}$: the PGL_3 pencil of these complexes is spectrally realizable by a non-negative integer matrix, irreducible of period 3.

Proof. (i) is computation (Table 1, line (R)); the grading statement is Theorem 4.1. (ii) The Boyle–Handelman conditions for a primitive integer matrix are: Λ_Q is the root multiset of a monic integer polynomial (Q), a simple strictly dominant positive root (q^6 , by (i)), and $\mathrm{tr}_m(\Lambda_Q) \geq 0$ for all m ; [4] proves them sufficient. (iii) If $M = \begin{pmatrix} 0 & X & 0 \\ 0 & 0 & Y \\ Z & 0 & 0 \end{pmatrix}$ then $M^3 = \mathrm{diag}(XYZ, YZX, ZXY)$, so the nonzero spectrum of M consists of all cube roots of the nonzero spectrum of $XYZ = B_0$, i.e. of the ω -orbits of $\Lambda_Q^{1/3}$, which is the nonzero spectrum of $C^{(3)}$. $\square$

Lemma 4.6 (Tail bound). *Let $M < q^2$ bound the moduli of the genuinely nontrivial eigenvalues and $N = n - 1$. Then $\mathrm{tr}_m(\Lambda_Q) \geq q^{6m} + q^{3m} + 1 - NM^{3m} - \sum_{d \leq m/2} (q^{6d} + q^{3d} + 1 + NM^{3d})$, positive for all $m \geq m_0$ with m_0 computable from (n, q, M) ; for the five complexes $m_0 \leq 3$.*

Proof. $|p_d(\Lambda_Q) - (1 + q^{3d} + q^{6d})| \leq NM^{3d}$; the proper divisors of m are $\leq m/2$; the bound increases with m once q^{6m} dominates, which for $M = q$ and $n \leq 168$ happens by $m = 3$. $\square$

Remark 4.7 (What remains of the realization problem). Theorem 4.5 is about the nonzero spectrum. The realization problem proper — a non-negative integer matrix B shift equivalent over $\mathbb{Z}$ to $C^{(3)}$, so that $\mathrm{coker}(I - B) \cong \mathbb{Z} \oplus \mathrm{Jac}^H(Y)$ with the Hecke dynamics — asks for more than spectral data, and we do not solve it. What is established is that the one tool which closed it in rank one, Theorem [II, 1.6], is unavailable in rank two, and that the necessary conditions known to us are all satisfied. The problem is open and, as far as the trace method can see, unobstructed. Whether the double-loop mechanism of Section 5 can be upgraded from a K -theoretic to a dynamical realization is the natural next question, and [II, Thm. 1.6] no longer forbids it here.

5. SIGNED SHADOWS AND THE HIGHER KIRCHBERG ALGEBRAS

Lemma 5.1 (Signed shadow lemma). *Let $M \in M_N(\mathbb{Z})$ with $M_{ee} \geq 1$ for all e , and let $\mathcal{P} = \{(e, f) : e \neq f, M_{ef} > 0\}$. Define a non-negative integer matrix B on the vertex set $\{e\} \sqcup \{w_{ef} : (e, f) \in \mathcal{P}\} \sqcup \{e'\}$ by the arcs*

- $|M_{ef}|$ arcs $f \rightarrow e$ for each $e \neq f$ with $M_{ef} < 0$;
- for $(e, f) \in \mathcal{P}$: M_{ef} arcs $f \rightarrow w_{ef}$, one arc $w_{ef} \rightarrow e$, two loops at w_{ef} ;
- for each e : one arc $e \rightarrow e'$, $M_{ee} - 1$ arcs $e' \rightarrow e$, two loops at e' .

Then, with $X := (I - B^t)_{\mathrm{main}, \mathrm{aux}}$, $Y := (I - B^t)_{\mathrm{aux}, \mathrm{main}}$ and $U = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix}$, $V = \begin{pmatrix} I & 0 \\ Y & I \end{pmatrix}$,

$$U(I - B^t)V = M \oplus (-I_{|\mathcal{P}|+N}).$$

If the graph on $\{e\}$ with edges $\{e, f\}$ for $M_{ef} \neq 0$ is connected, B is irreducible with double-loop vertices, and $\mathcal{O}_B := C^*(\text{digraph of } B)$ is a unital Kirchberg algebra with $K_0(\mathcal{O}_B) \cong \text{coker}_{\mathbb{Z}} M$ and $K_1(\mathcal{O}_B) \cong \ker_{\mathbb{Z}} M$.

Proof. The auxiliary block of $I - B^t$ is $-I$ (each auxiliary vertex carries exactly two loops and no other arcs among auxiliaries), so the Schur complement of that block is $R + XY$ where R is the main block: diagonal 1, entry M_{ef} at negative positions, 0 elsewhere. The auxiliary w_{ef} contributes $X_{e,w}Y_{w,f} = (-1)(-M_{ef}) = M_{ef}$ at (e, f) and nothing else; the shadow e' contributes $X_{e,e'}Y_{e',e} = -(M_{ee} - 1)(-1) = M_{ee} - 1$ at (e, e) . Hence $R + XY = M$, and the displayed identity is the block factorization $\begin{pmatrix} R & X \\ Y & -I \end{pmatrix} = U^{-1}((R + XY) \oplus (-I))V^{-1}$. Irreducibility: every arc $f \rightarrow e$ or $f \rightarrow w_{ef} \rightarrow e$ is matched by the reverse connection through M_{fe} or through the shared vertex structure of a connected support graph, and every e' communicates with e in both directions; the double loops give the witness of [VI, Thm. 3.1]; K -theory is the Cuntz–Krieger formula transported by U, V . For $M = \Delta_p$ of a graph ($M_{ef} \leq 0$ off the diagonal) $\mathcal{P} = \emptyset$ and the construction is [VI, Def. 1.1] with $c = M_{ee} - 1$. $\square$

Theorem 5.2 (Higher shadow algebras). *For an $\tilde{A}_2$ complex Y let $\mathcal{O}(Y, \Delta_1) := \mathcal{O}_{B(\Delta_1)}$ and $\mathcal{O}(Y, \Delta^H) := \mathcal{O}_{B(\Delta^H)}$ be the algebras of Lemma 5.1 for $M = \Delta_1$ (diagonal $q + 3 \geq 1$) and $M = \Delta^H$ (diagonal $q^3 - 1 \geq 1$ for $q \geq 2$). Both are unital Kirchberg algebras, hence traceless, and*

$$K_0(\mathcal{O}(Y, \Delta_1)) \cong \mathbb{Z}^{b_1} \oplus \text{Jac}_1(Y), \quad K_1 \cong \mathbb{Z}^{b_1}; \quad K_0(\mathcal{O}(Y, \Delta^H)) \cong \mathbb{Z} \oplus \text{Jac}^H(Y), \quad K_1 \cong \mathbb{Z},$$

the second under the hypothesis of Theorem 3.2(ii). Type- and orientation-preserving automorphisms of Y act compatibly. Changing the edge orientations conjugates Δ_1 by a diagonal ± 1 matrix and changes $B(\Delta_1)$, but not the K -theory.

Proof. Lemma 5.1 with Proposition 2.2 and Theorem 3.2; the support graph of Δ_1 is connected because two edges of a connected complex are joined by a chain of edges sharing vertices, and pairs spanning a triangle (where the entry is 0) are joined through a third edge; the support of Δ^H is the 1-skeleton. $\square$

Remark 5.3 (Digraph, not cellular algebra). The specification asked for a “two-dimensional shadow complex with higher self-loop faces” and a “higher graph or cellular C^* -algebra”. No such theory is needed: the invariant to be realized is a cokernel, cokernels are unimodular-equivalence invariants, and Lemma 5.1 realizes any integer matrix with positive diagonal as the Bowen–Franks presentation of an ordinary graph algebra. The two dimensions of Y enter through Δ_1 and Δ^H , not through the algebra.

6. THE VERIFICATION BATTERY

All computations are exact except the spectra. Smith forms are computed p -adically: the determinant of the (reduced) matrix by Bareiss elimination, its prime factors by trial division, and the p -primary invariants by elimination over $\mathbb{Z}/p^k$ with valuation-minimal pivots; for singular symmetric M the torsion found is certified complete by the lattice identity

$$(6.1) \quad |\text{Tor coker}_{\mathbb{Z}} M| \cdot \text{disc}(\ker_{\mathbb{Z}} M) = \text{pdet}(M),$$

where disc is the Gram determinant of a saturated basis of the kernel lattice and pdet the product of the nonzero eigenvalues; the two sides agree to 10^{-6} in every case (this also verifies (6.1) itself, a matrix-tree theorem for symmetric integer matrices, on twelve instances).

Data. Y_{168} : $n = 168$, $|E| = |F| = 1176$, $\chi = 168$, $(b_0, b_1, b_2) = (1, 0, 167)$, flag condition holds;

$$\text{Jac}_0 = (\mathbb{Z}/2)^{31} \oplus (\mathbb{Z}/16)^6 \oplus (\mathbb{Z}/592)^6 \oplus (\mathbb{Z}/1184)^{18} \oplus \mathbb{Z}/4736 \\ \oplus (\mathbb{Z}/33152)^3 \oplus (\mathbb{Z}/430976)^{12} \oplus \mathbb{Z}/3016832 \oplus \mathbb{Z}/9050496,$$

$$\text{Jac}^H = (\mathbb{Z}/7)^{18} \oplus (\mathbb{Z}/299593)^{13} \oplus \mathbb{Z}/2097151 \oplus \mathbb{Z}/14680057 \oplus (\mathbb{Z}/117440456)^5 \oplus \mathbb{Z}/352321368,$$

where $299593 = 7 \cdot 127 \cdot 337$ and $2097151 = 2^{21} - 1 = 7^2 \cdot 127 \cdot 337$; $|\text{Jac}^H| \approx 6.18 \cdot 10^{136}$ agrees with $\frac{1}{n} \prod_j |P_j(1)|$ to 10^{-14} ; $\text{Tr}(C^{(3)})^3 = 504 = n(q-1)^2(q+1)$.

Y_{21} : $(b_0, b_1, b_2) = (1, 0, 20)$. $\text{Jac}_0 = \mathbb{Z}/2 \oplus (\mathbb{Z}/14)^{15} \oplus \mathbb{Z}/98 \oplus \mathbb{Z}/294$; $\text{Jac}_1 = (\mathbb{Z}/2)^{10} \oplus (\mathbb{Z}/14)^{10} \oplus (\mathbb{Z}/98)^{26} \oplus \mathbb{Z}/2058 \oplus \mathbb{Z}/18522$ ($|\text{Jac}_1| \approx 6.7 \cdot 10^{73}$); $\text{Jac}_2 = (\mathbb{Z}/2)^{13} \oplus (\mathbb{Z}/14)^{15} \oplus \mathbb{Z}/98 \oplus \mathbb{Z}/294$; $K(Y) = (\mathbb{Z}/7)^6 \oplus \mathbb{Z}/14 \oplus (\mathbb{Z}/98)^{10} \oplus \mathbb{Z}/294$; $\text{Jac}^H = (\mathbb{Z}/7)^{17} \oplus \mathbb{Z}/147$, $|\text{Jac}^H| = 3 \cdot 7^{19}$; $\text{Tr}(C^{(3)})^3 = 651 = 63 + 588$.

Y_{24} : $(1, 0, 23)$. $\text{Jac}_0 = (\mathbb{Z}/13)^7 \oplus \mathbb{Z}/26 \oplus (\mathbb{Z}/208)^4 \oplus \mathbb{Z}/1456 \oplus \mathbb{Z}/4368$; $\text{Jac}_1 = (\mathbb{Z}/2)^{10} \oplus (\mathbb{Z}/51376)^2 \oplus (\mathbb{Z}/102752)^6 \oplus (\mathbb{Z}/2877056)^4 \oplus \mathbb{Z}/60418176 \oplus \mathbb{Z}/543763584$; $\text{Jac}_2 = \mathbb{Z}/2 \oplus (\mathbb{Z}/26)^6 \oplus (\mathbb{Z}/52)^2 \oplus (\mathbb{Z}/208)^2 \oplus (\mathbb{Z}/1456)^3 \oplus \mathbb{Z}/4368$; $K(Y) = (\mathbb{Z}/2)^3 \oplus (\mathbb{Z}/38)^7 \oplus \mathbb{Z}/152 \oplus (\mathbb{Z}/304)^2 \oplus (\mathbb{Z}/2128)^3 \oplus \mathbb{Z}/6384$; $\text{Jac}^H = (\mathbb{Z}/7)^2 \oplus \mathbb{Z}/49 \oplus (\mathbb{Z}/392)^5 \oplus \mathbb{Z}/1176$.

Y_{42} : $(1, 0, 41)$. $\text{Jac}_0 = (\mathbb{Z}/4)^2 \oplus (\mathbb{Z}/28)^{10} \oplus \mathbb{Z}/56 \oplus (\mathbb{Z}/8288)^4 \oplus \mathbb{Z}/754208 \oplus \mathbb{Z}/2262624$;

$$\text{Jac}_1 = (\mathbb{Z}/2)^{12} \oplus (\mathbb{Z}/4)^4 \oplus (\mathbb{Z}/28)^{10} \oplus (\mathbb{Z}/196)^{14} \oplus (\mathbb{Z}/784)^6 \oplus (\mathbb{Z}/1568)^2 \\ \oplus (\mathbb{Z}/21865186112)^4 \oplus \mathbb{Z}/N_1 \oplus \mathbb{Z}/N_2,$$

$N_1 = 1474391364718272$, $N_2 = 13269522282464448 = 9N_1$; $\text{Jac}_2 = (\mathbb{Z}/2)^6 \oplus (\mathbb{Z}/4)^6 \oplus (\mathbb{Z}/28)^{10} \oplus \mathbb{Z}/56 \oplus (\mathbb{Z}/8288)^4 \oplus \mathbb{Z}/754208 \oplus \mathbb{Z}/2262624$; $K(Y) = \mathbb{Z}/7 \oplus (\mathbb{Z}/14)^6 \oplus (\mathbb{Z}/98)^5 \oplus \mathbb{Z}/1078 \oplus (\mathbb{Z}/2156)^3 \oplus \mathbb{Z}/40964 \oplus \mathbb{Z}/245784$; $\text{Jac}^H = (\mathbb{Z}/7)^{18} \oplus \mathbb{Z}/299593 \oplus \mathbb{Z}/2097151 \oplus \mathbb{Z}/176160684$.

Y_{84} (vertex level): $\text{Jac}_0 = (\mathbb{Z}/2)^{12} \oplus (\mathbb{Z}/4)^7 \oplus (\mathbb{Z}/16)^2 \oplus \mathbb{Z}/1184 \oplus (\mathbb{Z}/8288)^5 \oplus (\mathbb{Z}/16576)^6 \oplus (\mathbb{Z}/215488)^4 \oplus \mathbb{Z}/1508416 \oplus \mathbb{Z}/4525248$.

Thin torus $m = 6$ ($n = 36$, $|E| = 108$, $|F| = 72$, $\chi = 0$, $(1, 2, 1)$, flag holds): $\text{Jac}_0 = \mathbb{Z}/2 \oplus (\mathbb{Z}/8)^2 \oplus (\mathbb{Z}/16)^6 \oplus \mathbb{Z}/80 \oplus (\mathbb{Z}/240)^3 \oplus \mathbb{Z}/720 \oplus \mathbb{Z}/2160$; $\text{coker } \Delta_1 = \mathbb{Z}^2 \oplus (\mathbb{Z}/2)^{12} \oplus \mathbb{Z}/8 \oplus (\mathbb{Z}/16)^3 \oplus (\mathbb{Z}/640)^3 \oplus (\mathbb{Z}/1920)^5 \oplus (\mathbb{Z}/5760)^2 \oplus \mathbb{Z}/17280 \oplus \mathbb{Z}/51840$; $K(Y) = \mathbb{Z}^2 \oplus \mathbb{Z}/72$; $\text{Tr}(C^{(3)})^3 = 0$. The torus $m = 3$ violates the flag condition ($\text{Tr } A_1^3 = 81 > 54$) and has $\text{Tr}(C^{(3)})^3 = 27$.

In every thick example the odd parts of Jac_0 and Jac_2 nearly coincide while the 2-parts differ, $K(Y)$ is much smaller than Jac_1 , and Jac^H shares no visible structure with any Hodge group; we record this without interpretation.

check	complexes	result
(P) presentation axioms; simplicial; Heawood/hexagon links; (ii),(iii); flag (iv)	all	exact; (iv) only for Y_{168} , T_6
(D) $\partial_1 \partial_2 = 0$; Betti numbers; χ ; $b_2 = \chi - 1 + b_1$	all	exact
(H) Smith forms of $\Delta_0, \Delta_1, \Delta_2$; $K(Y)$; completeness via (6.1)	thin, Y_{21}, Y_{24}, Y_{42}	exact, complete
(L) Δ^H : two-sided kernel; equal cofactors; $ \text{Jac}^H = \frac{1}{n} \prod P_j(1) $	thick	exact; numeric to 10^{-14}
(C) $U(I - C^{(3)})V = P(1) \oplus I \oplus I$, $P(1) = -\Delta^H$	all	exact
(T) $\text{Tr}(C^{(3)})^k$, $k \leq 6$, directly and by Newton; $= 0$ for $3 \nmid k \leq 30$; $\text{Tr}(C^{(3)})^3$ formula	all	exact
(R) nine trivial eigenvalues; nontrivial $ \lambda = q$ (Ramanujan); Λ_Q net traces $m \leq 10$; m_0	thick	numeric / exact
(S) signed shadows of Δ^H and Δ_1 : identity, $B \geq 0$, irreducible, double loops; Smith check	all	exact

TABLE 1. Exact verification (pgl3_building.py). Sizes: shadow of Δ^H has $3n + |E|$ vertices (1512 for Y_{168}); shadow of Δ_1 has 2058 vertices for Y_{21} and 2352 for Y_{24} . Net traces of Λ_Q for Y_{168} : 168, 3920, 257544, 16751728, $\dots$; $m_0 = 3$.

7. SCOPE, RESIDUE, CORRECTIONS

Remark 7.1 (Solved, open, impossible). *Solved*: the Hodge and Hecke Laplacians of $\tilde{A}_2$ complexes are set up and their critical groups computed exactly on thin and thick examples; the Hecke critical group is identified with the product of inverse GL_3 L -values and the companion is reduced to it unimodularly; the trace question is settled, negatively for the obstruction and positively for spectral realizability on the computed complexes; the signed-shadow lemma gives Kirchberg realizations of Δ_1 and Δ^H . *Open*: (a) a non-negative matrix shift equivalent over $\mathbb{Z}$ to $C^{(3)}$ (Remark 4.7); (b) a general theorem that all net traces of the reduced pencil are non-negative for every $\tilde{A}_2$ complex of order $q \geq 2$ (we have it for five complexes, and in general for the Ramanujan case once $m \geq m_0$); (c) a relation between $\mathrm{Jac}_1(Y)$, $K(Y)$ and the Hecke data — none is visible in the computed examples, and we conjecture none exists: the edge critical group is combinatorial, the vertex Hecke group arithmetic; (d) towers of $\tilde{A}_2$ complexes (the natural continuation of [III]–[IX]). *Impossible or false as stated*: a cubic trace obstruction (Theorem 4.2).

Remark 7.2 (Corrections to the task specification). (1) The predicted “higher cubic trace obstruction” does not exist: $\mathrm{Tr}(C^{(3)})^k = 0$ for $3 \nmid k$ and $\mathrm{Tr}(C^{(3)})^3 = n(q-1)^2(q+1) \geq 0$; the paper proves the contrary and, for the computed complexes, spectral realizability. (2) The “duality bridge between the cellular sandpile group and the GL_3 local L -factors” runs through the Hecke–Laplacian Δ^H and its critical group $\mathrm{Jac}^H(Y)$, not through the Hodge groups $\mathrm{Jac}_0, \mathrm{Jac}_1$, which in rank two are different objects (Theorem 3.2(iii)). (3) The test complexes are $\tilde{A}_2$ complexes of order 2 built from a Cartwright–Steger-type triangle presentation, i.e. quotients of an $\tilde{A}_2$ building of order 2; the specification’s “ $\mathrm{PGL}_3(\mathbb{Q}_p)$ ” is not literally available (arithmetic lattices in $\mathrm{PGL}_3(\mathbb{Q}_p)$ were not constructed), and nothing proved depends on the field. (4) No “two-dimensional shadow complex” is needed (Remark 5.3). (5) The companion in the specification is correct, and its reduction to Δ^H (Theorem 3.3) is the rank-two form of [II, §1.1].

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Unrefereed preprint. Every numbered statement is proved from finite linear algebra and the results of Papers I–IX; the classical inputs are [1] (existence of the building for a triangle presentation), the Cuntz–Krieger facts quoted in [VI], and [4] for Theorem 4.5 only. Spectral statements marked “numeric” in Table 1 are floating-point computations with the stated tolerances; all group-theoretic and trace statements are exact. The identification of the thick complexes with quotients of the Bruhat–Tits building of PGL_3 over a specific local field is not claimed. Bibliographic data of [1, 3, 4, 2] were checked against publisher or indexing pages.

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